

The Spectrogram Associated with the Linear Canonical Transform

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Abstract—linear canonical transform is a generalized form of Fourier Transform, and it has big effect in modern applied mathematics and signal processing community, especially in detection and parameter estimation of non-stationary signal. We investigate the spectrogram associated with the linear canonical transform in this paper, a novel definition of spectrogram is proposed and the basic properties are deduced accordingly. Finally, the simulations are also given to test the conclusions which we have obtained.

Keywords—linear canonical transform; spectrogram; linear frequency modulation signal(LFM); parameter estimation

I. INTRODUCTION

With the research progress of the fractional Fourier transform in the field of signal processing in recent years [1], the research of the linear canonical transform is becoming one of the hottest research topics [1-6], the linear canonical transform (LCT) is a generalized form of the Fourier transform and the fractional Fourier transform, it has achieved excellent results in practical application. These researches show that the theory of linear canonical transform is one of the most important methods of non-stationary signal processing further. Therefore it has the great significance for further completing the LCT's theoretical system and advancing applications and the developments in the field of mathematics and signal processing to study the related LCT theory.

At the same time, the spectrogram [7] is one of the important conceptions in signal processing community. It is the key point in the energy distribution and estimation of signal. For the best of our knowledge, there no paper has been published associate with the spectrogram in LCT domain. So it is interesting and worthwhile to investigate the definition and properties of the spectrogram in the LCT domain.

Focus on this problem this paper generalize the classical spectrogram to the LCT domain, the definition, properties and simulation results are also derived according.

II. PRELIMINARY

A. Spectrogram:

The spectrogram of signal $s(t)$ is defined as following [7]:

$$P(t, \omega) = \left| \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} s(\tau) h(\tau - t) e^{-j\omega\tau} d\tau \right|^2 \quad (1)$$

where $h(t)$ is a window function.

The spectrogram is the energy density spectrum. By choosing a window that weighs the interval near the observation point a greater amount than other points, the spectrogram can be used to estimate local quantities [7-9].

B. The Linear Canonical Transform (LCT)

The LCT of a signal $s(t)$ is defined as [2]

$$F_{(a,b,c,d)}(u) = \begin{cases} \int_{-\infty}^{+\infty} s(t) K_{(a,b,c,d)}(t, u) dt, & b \neq 0 \\ \sqrt{d} e^{j\frac{cd}{2}u^2} s(du), & b = 0 \end{cases} \quad (2)$$

where $K_{(a,b,c,d)}(t, u) = \sqrt{\frac{1}{j2\pi b}} e^{j(\frac{d}{2b}u^2 - \frac{1}{b}ut + \frac{a}{2b}t^2)}$ is the kernel function, and the parameters (a, b, c, d) can be expressed as parameter matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, which must satisfy $ad - bc = 1$, i.e. $|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = 1$.

Consideration of convenience for writing, we use $L_{(a,b,c,d)}(\bullet)$ to represent the operator LCT with the parameter (a, b, c, d) . Based on properties of the LCT, the classical time-frequency analysis methods have been generalized to the LCT domain [10-19].

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III. THE SPECTROGRAM ASSOCIATED WITH THE LINEAR CANONICAL TRANSFORM

A. Definition

Compare with the traditional definition of spectrogram, we give a novel definition of the spectrogram associated with the linear canonical transform as the followings:

$$P_A^s(t, u) = \left| \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} s(\tau) h(\tau - t) K_A(\tau, u) d\tau \right|^2 \quad (3)$$

where $K_A(t, u)$ is the kernel function of LCT.

B. Properties of the spectrogram associated with the linear canonical transform

1) Real property:

$$P_A^s(t, u) = (P_A^s(t, u))^* \quad (4)$$

2) Shifting properties:

a) time shifting: If $s(t) \rightarrow s(t - t_0)$, then

$$P_A^s(t, u) \rightarrow \left| e^{-j\frac{1}{b}ut_0} \right|^2 P_A^{s(t)}(t - t_0, u) \quad (a = 0) \quad (5)$$

b) frequency shifting: If $s(t) \rightarrow e^{j\omega_0 t} s(t)$, then

$$P_A^s(t, u) \rightarrow P_A^{s(t)}(t, u - b\omega_0) \quad (d = 0) \quad (6)$$

c) If $s(t) \rightarrow e^{j\omega_0 t^2} s(t)$, then

$$P_A^{s(t)}(t, u) \rightarrow P_B^{s(t)}(t, u) \quad (7)$$

$$\text{where } B = \begin{bmatrix} a + 2b\omega_0 & b \\ c & d \end{bmatrix}, \text{ and } |B| = 1.$$

d) If $s(t) \rightarrow e^{j\omega_0 t} s(t - t_0)$, then

$$P_A^{s(t)}(t, u) \rightarrow \left| e^{-j\frac{1}{b}(u - b\omega_0)t_0} \right|^2 P_A^{s(t)}(t - t_0, u - b\omega_0), (a = d = 0) \quad (8)$$

3) Limited time-delay support

If $s(t) = 0, (t \notin [t_1, t_2])$, then

$$P_A^s(t, u) = 0 (t \notin [t_1, t_2]) \quad (9)$$

4) Group delay:

$$\tau_s(u) = \frac{\int_{-\infty}^{+\infty} t P_A^s(t, u) du}{\int_{-\infty}^{+\infty} P_A^s(t, u) du} \quad (10)$$

5) Anti-derivative property:

$$s(t) = \frac{2\pi}{s^*(0)h(0)h^*(-t)} \int_{-\infty}^{+\infty} P_A^s(t, u) K_{A^{-1}}(t, u) K_{B^{-1}}(0, u) du \quad (11)$$

where $A^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, $B^{-1} = \begin{bmatrix} d & b \\ c & a \end{bmatrix}$, $h(t)$ is the window function.

Proof: According to (3), we can see that

$$\begin{aligned} P_A^s(t, u) &= \left| \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} s(\tau) h(\tau - t) K_A(\tau, u) d\tau \right|^2 \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} s(\tau) h(\tau - t) K_A(\tau, u) d\tau \times \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} s(\tau') h(\tau' - t) K_A(\tau', u) d\tau' \right)^* \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} s(\tau) s^*(\tau') h(\tau - t) h^*(\tau' - t) K_A(\tau, u) (K_A(\tau', u))^* d\tau d\tau' \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} s(\tau) s^*(\tau') h(\tau - t) h^*(\tau' - t) K_A(\tau, u) K_B(\tau', u) d\tau d\tau' \end{aligned}$$

$$\text{where } B = \begin{bmatrix} a & -b \\ -c & d \end{bmatrix}.$$

Then based on the inverse linear canonical transform (ILCT), we get:

$$\begin{aligned} &s(\tau) s^*(\tau') h(\tau - t) h^*(\tau' - t) \\ &= 2\pi \int_{-\infty}^{+\infty} P_A^s(t, u) K_{A^{-1}}(\tau, u) K_{B^{-1}}(\tau', u) du \end{aligned}$$

$$\text{where } A^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, B^{-1} = \begin{bmatrix} d & b \\ c & a \end{bmatrix}.$$

Let $\tau = t, \tau' = 0$, then we can get:

$$s(t) = \frac{2\pi}{s^*(0)h(0)h^*(-t)} \int_{-\infty}^{+\infty} P_A^s(t, u) K_{A^{-1}}(t, u) K_{B^{-1}}(0, u) du$$

where $h(t)$ is the window function. $\square$

6) Marginal properties of time-delay:

$$P_A^s(0, u) = \left| \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} s(\tau) h(\tau) K_A(\tau, u) d\tau \right|^2 \quad (12)$$

7) Marginal properties of frequency-shift:

$$P_A^s(t, 0) = \left| \frac{1}{2\pi\sqrt{bj}} \int_{-\infty}^{+\infty} s(\tau) h(\tau - t) e^{j\frac{a}{2b}\tau^2} d\tau \right|^2 \quad (13)$$

$$\text{Proof: } P_A^s(t, 0) = \left| \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} s(\tau) h(\tau - t) K_A(\tau, 0) d\tau \right|^2$$

$$\text{hence } K_A(\tau, 0) = \sqrt{\frac{1}{j2\pi b}} e^{j\frac{a}{2b}\tau^2}$$

$$\text{so } P_A^s(t, 0) = \left| \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} s(\tau) h(\tau - t) \sqrt{\frac{1}{j2\pi b}} e^{j\frac{a}{2b}\tau^2} d\tau \right|^2$$

$$= \left| \frac{1}{2\pi\sqrt{bj}} \int_{-\infty}^{+\infty} s(\tau) h(\tau - t) e^{j\frac{a}{2b}\tau^2} d\tau \right|^2 \quad \square$$

8) Scaling property:

If $s(t) = \sqrt{|\lambda|} f(\lambda t)$, then

$$P_A^s(t, u) = \left| \sqrt{|\lambda|} \right|^2 P_{1B}^f(\lambda t, \frac{u}{\lambda}) \quad (14)$$

where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B = \begin{bmatrix} a/\lambda^2 & b \\ c & \lambda^2 d \end{bmatrix}$, $P_{1B}^f(\lambda t, \frac{u}{\lambda})$ is

LCTS, whose window function is $h_1(t) = h(t/\lambda)$, and parameter matrix is B . And $h_1(t) = h(t/\lambda)$.

Proof:

$$P_A^s(t, u) = \left| \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} s(\tau) h(\tau - t) K_A(\tau, u) d\tau \right|^2$$

$$= \left| \frac{\sqrt{|\lambda|}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(\lambda\tau) h(\tau - t) K_A(\tau, u) d\tau \right|^2$$

$$\text{And } K_A(\tau, u) = \sqrt{\frac{1}{j2\pi b}} e^{j(\frac{d}{2b}u^2 - \frac{1}{b}u\tau + \frac{a}{2b}\tau^2)}$$

$$= \sqrt{\frac{1}{j2\pi b}} e^{j\left[\frac{\lambda^2 d}{2b}\left(\frac{u}{\lambda}\right)^2 - \frac{1}{b}\left(\frac{u}{\lambda}\right)\lambda\tau + \frac{\left(\frac{a}{\lambda^2}\right)}{2b}(\lambda\tau)^2\right]}$$

$$= K_B(\lambda\tau, u/\lambda)$$

$$\text{where } B = \begin{bmatrix} a/\lambda^2 & b \\ c & \lambda^2 d \end{bmatrix}.$$

Then we get

$$P_A^s(t, u) = \left| \frac{\sqrt{|\lambda|}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(\lambda\tau) h(\tau - t) K_A(\tau, u) d\tau \right|^2$$

$$= \left| \frac{\sqrt{|\lambda|}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(\lambda\tau) h(\tau - t) K_B(\lambda\tau, u/\lambda) d\tau \right|^2$$

$$= \left| \frac{\sqrt{|\lambda|}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(\lambda\tau) h_1(\lambda\tau - \lambda t) K_B(\lambda\tau, u/\lambda) d\tau \right|^2$$

$$= \left| \sqrt{|\lambda|} \right|^2 P_{1B}^f(\lambda t, \frac{u}{\lambda})$$

where $P_{1B}^s(t, u)$ is LCTS, whose parameter matrix is

$$B = \begin{bmatrix} a/\lambda^2 & b \\ c & \lambda^2 d \end{bmatrix}, \text{ window function is } h_1(t) = h(t/\lambda).$$

□

9) Relationship between LCTS and the classical spectrogram

If $A = (a, b; c, d) = (0, 1; -1, 0)$, then

$$P_{(0,1;-1,0)}^s(t, u) = \frac{1}{4\pi^2} P(t, u) \quad (15)$$

where $P(t, u)$ is the classical spectrogram of signal $s(t)$.

$$\text{Proof: } K_{(0,1;-1,0)}(\tau, u) = \sqrt{\frac{1}{j2\pi}} e^{-ju\tau}$$

$$P_{(0,1;-1,0)}^s(t, u) = \left| \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} s(\tau) h(\tau - t) \sqrt{\frac{1}{j2\pi}} e^{-ju\tau} d\tau \right|^2$$

$$= \frac{1}{4\pi^2} \left| \int_{-\infty}^{+\infty} s(\tau) h(\tau - t) e^{-ju\tau} d\tau \right|^2$$

$$= \frac{1}{4\pi^2} P(t, u)$$

where $P(t, u)$ is the classical spectrogram of signal $s(t)$. □

IV. THE SPECTROGRAM OF LINEAR FREQUENCY MODULATION SIGNAL

A. Corollary

The LCTS of LFM signal $s(t) = e^{j(\omega_0 t + m t^2/2)}$ is

$$P_A^s(t, u) = \left| \sqrt{\frac{b}{j}} e^{j\frac{d}{2b}u^2} \delta(u - b\omega_0) \right|^2, \frac{a}{b} + m = 0 \quad (16)$$

where the parameters ω_0, m are initial phase and frequency fluctuation rate of signal $s(t)$. And the window function is $h(t) \equiv 1$.

Proof:

According to (3), The LCTS of LFM signal $s(t) = e^{j(\omega_0 t + m t^2/2)}$ is

$$P_A^s(t, u) = \left| \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} s(\tau) h(\tau - t) K_A(\tau, u) d\tau \right|^2$$

$$= \left| \frac{1}{2\pi\sqrt{bj}} \int_{-\infty}^{+\infty} h(\tau - t) e^{j[\frac{d}{2b}u^2 - \frac{1}{b}(u - b\omega_0)\tau + (\frac{a}{2b} + \frac{m}{2})\tau^2]} d\tau \right|^2$$

$$= \left| \frac{1}{2\pi\sqrt{bj}} e^{j\frac{d}{2b}u^2} \int_{-\infty}^{+\infty} h(\tau - t) e^{j[(\frac{a}{2b} + \frac{m}{2})\tau^2 - \frac{1}{b}(u - b\omega_0)\tau]} d\tau \right|^2$$

Additionally, $h(t) \equiv 1$, so

$$P_A^s(t, u) = \left| \frac{1}{2\pi\sqrt{bj}} e^{j\frac{d}{2b}u^2} \int_{-\infty}^{+\infty} e^{j[(\frac{a}{2b} + \frac{m}{2})\tau^2 - \frac{1}{b}(u - b\omega_0)\tau]} d\tau \right|^2 \quad (17)$$

When $\frac{a}{b} + m = 0$, (17) is

$$P_A^s(t, u) = \left| \frac{1}{2\pi\sqrt{bj}} e^{j\frac{d}{2b}u^2} \int_{-\infty}^{+\infty} e^{-j\frac{1}{b}(u-b\omega_0)\tau} d\tau \right|^2$$

$$= \left| \sqrt{\frac{b}{j}} e^{j\frac{d}{2b}u^2} \delta(u - b\omega_0) \right|^2 \quad \square$$

B. Simulation

Choose bounded LFM signal

$$s(t) = e^{j\left(\omega_0 t + \frac{m}{2}t^2\right)}, |t| < \frac{T}{2} \quad (18)$$

where $T = 40, \omega_0 = 10, m = 0.8$,

$A = (a, b; c, d) = (0.8, -1; 0, 1.25)$, so according to (16), the LCTS of the signal (18) is

$$P_A^s(t, u) = \left| \sqrt{j} e^{-j0.625u^2} \delta(u + 10) \right|^2 \quad (19)$$

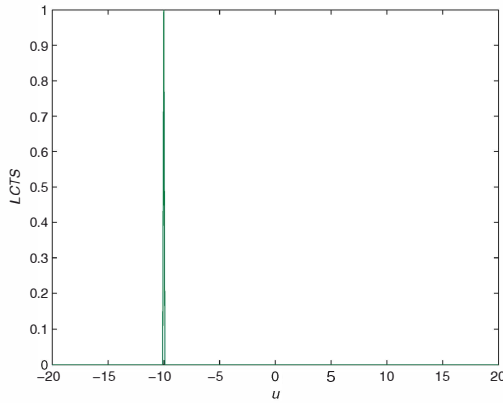


Fig. 1. The LCTS of signal (18)'s LCTS in u .

The LCTS of signal in equation 18 was plotted in Figure 1 using Matlab [20], as can be seen from the figure 1, the LCTS of signal $s(t)$ doesn't value 0, when the image of $u(t)$ trend as a specific line. In addition, according to (19), we can see that when $u = -b\omega_0 = -10$, there is an impulse. This is consistent with the situation of Figure 1. So on the other hand we can get the initial phase ω_0 by the LCTS of signal (18)'s LCTS in u .

V. CONCLUSION

The spectrogram plays an important role on the description, analysis, detection and estimation of the signal energy. The basic properties of the spectrogram have been a hot point in the modern practical mathematic. In this paper, on the foundation of the theory, which we already know, we put forward the definition of the LCTS and study the properties of the new

definition. Actually, the LCTS is a father generation of the classical spectrogram. It is more useful in some field, especially in detection and estimation of signal. The research results have enriched the theory of LCT further, and it has showed that it will have excellent effect in practical engineering application in the future.

REFERENCES

- [1] Ran Ran, Bing Deng, and Yue Wang, Fractional Fourier Transform and Its Applications, Beijing: Tsinghua University Press, 2009
- [2] Tian-Zhou Xu, Bing-Zhao Li, Linear Canonical Transform and Its Applications, Beijing: Science Press, 2013
- [3] Bing-Zhao Li, Ran Tao, Yue Wang, "New sampling formulae related to linear canonical transform," Signal Processing, 2007, 87: 983-990. □
- [4] Bing-Zhao Li, Tian-Zhou Xu. "Sampling in the linear canonical transform domain," Mathematical Problems in Engineering, 2012, 2012: Article ID 504580, 1-13.
- [5] Bing Deng, Ran Tao, "Convolution theorems for the linear canonical transform and their applications," Science in China Series F: Information Sciences, 2006, 49(5): 592-603.
- [6] Cui-Ping Li, Bing-Zhao Li, Tian-Zhou Xu, "Approximating bandlimited signals associated with the LCT domain from nonuniform samples at unknown locations," Signal Processing, 2012, 92(7): 1658-1664.
- [7] L. Cohen, "Time-Frequency Distribution-A Review," Proceedings of the IEEE, vol. 77, pp. 941-981, July 1989.
- [8] J. G. Kirkwood, "Quantum statistics of almost classical assemblies," Physical Review, vol. 44, pp. 31-37, July 1933.
- [9] W. Rihaczek, "Signal energy distribution in time and frequency," IEEE Trans. Informat. Theory, vol. IT-14, pp. 369-374, May 1968.
- [10] Xiang Qiang, Qin Kai-yu, "A Time-frequency Analysis Method Based on Linear Canonical Transform and Short-time Fourier Transform," Acta Electronica Sinica, vol. 39, pp. 1508-1513, July 2011.
- [11] L. Cohen, Bai Ju-Xian, Time-Frequency Analysis: Theory and Applications, Xi'an: Xi'an Jiaotong University Press, 1988, pp. 94-109
- [12] Rui-Feng Bai, Bing-Zhao Li, "Wigner-Ville Distribution Associated with the Linear Canonical Transform", Journal of Applied Mathematics, vol. 2012, Article ID 740161, pp.1-14, May 2012
- [13] Tian-Wen Che, Bing-Zhao Li, Tian-Zhou Xu, "The ambiguity function associated with the linear canonical transform," EURASIP Journal on Advances in Signal Processing, vol. 2012(138), pp.1-14.
- [14] Guan-Lei Xu, Xiao-Tong Wang, Xiao-Gang Xu, "Generalized Hilbert transform and its properties in 2D LCT domain," Signal Processing, 2009, 89(7): 1395-1402.
- [15] Bing-Zhao Li, Tian-Zhou Xu, "Spectral analysis of sampled signals in the linear canonical transform domain," Mathematical Problems in Engineering, 2012, 2012: Article ID 536464, 1-19.
- [16] Juang Zhao, Ran Tao, Yue Wang, "Multi-channel filter banks associated with linear canonical transform," Signal Processing, 2013, 93: 695-705.
- [17] Hui Zhao, Qi-Wen Ran, Jing Ma, "Linear canonical ambiguity function and linear canonical transform moments," Optik, 2011, 122(6): 540-543.
- [18] Ran Tao, Yu-E Song, Yue Wang, "Ambiguity function based on the linear canonical transform," IET Signal Processing, 2012, 6(6): 568-576.
- [19] Jun-Ping Yan, Bing-Zhao Li, Yi-Hong Chen, "Wigner distribution moments associated with the linear canonical transform," International Journal of Electronics, vol.100, no. 4, pp.473-481.
- [20] Holly Moore, Gao Hui-Sheng, Matlab Practical Guide (Second Edition), Beijing: Electronic Industry Press, 2011